

AI4Chemical Sciences Bootcamp — Introduction

An introduction to AI for the Chemical Sciences — Bootcamp overview



J. Schleinitz · Monday, August 10, 2026 · RSC 275, Caltech

Morning Agenda

- I. **AI & ML today**
- II. Bootcamp Overview
- III. Bootcamp Objectives & Roadmap
- IV. Logistics

Artificial Intelligence – Definition

AI is the capability of computational systems to perform tasks typically associated with human intelligence, such as:

- Learning
- Reasoning
- Planning
- Perception
- Communication
- ...



Artificial Intelligence Forecasts?

When will there be a 50% chance that Human-level Artificial Intelligence exists?

Our World
in Data

Timelines of 812 AI experts surveyed in three studies between 2018 and 2022.

1) Timelines of 356 AI experts, surveyed in 2022 by Katja Grace et al.:

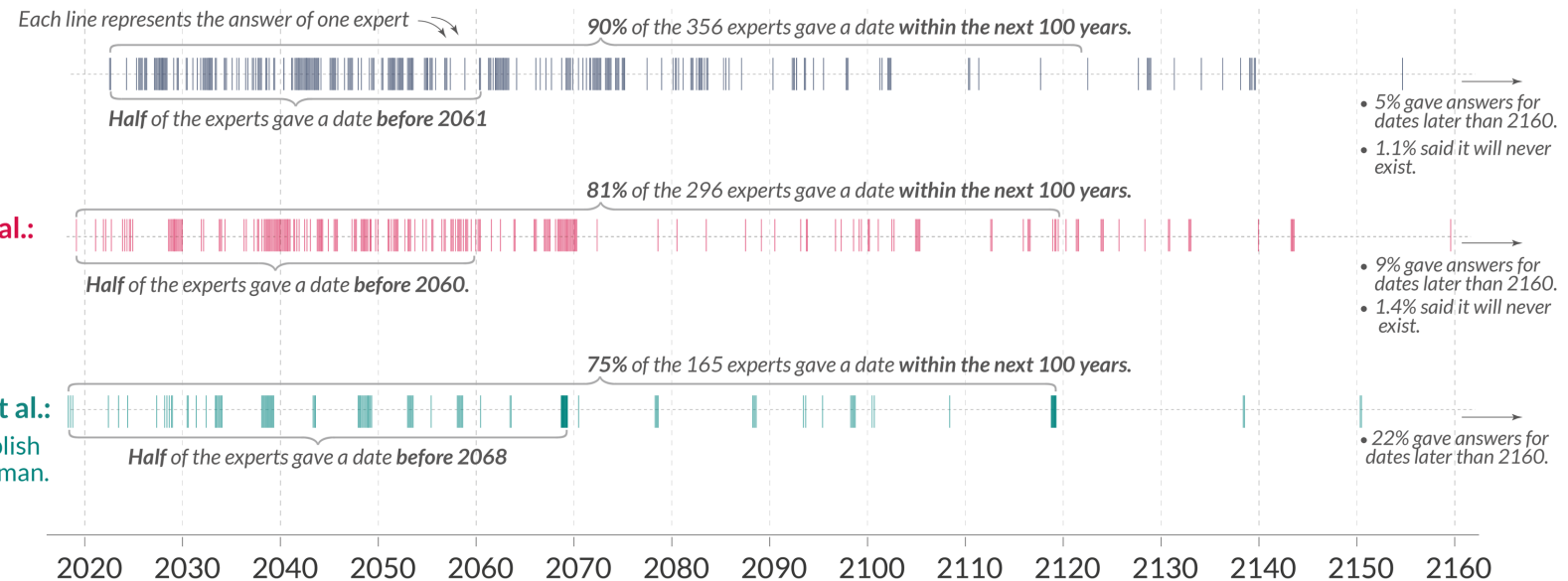
The experts were asked when unaided machines will be able to accomplish every task better and more cheaply than human workers.

2) Timelines of 296 AI experts, surveyed in 2019 by Baobao Zhang et al.:

The experts were asked when machines will collectively be able to perform more than 90% of all tasks that are economically relevant better than the median human paid to do that task.

3) Timelines of 165 AI experts, surveyed in 2018 by Gruetzemacher et al.:

The experts were asked when AI systems will collectively be able to accomplish 99% of tasks that humans are paid to do at or above the level of a typical human.



Full details on all studies and the questions that the AI experts were asked can be found in the text at [OurWorldInData.org/ai-timelines](https://ourworldindata.org/ai-timelines).

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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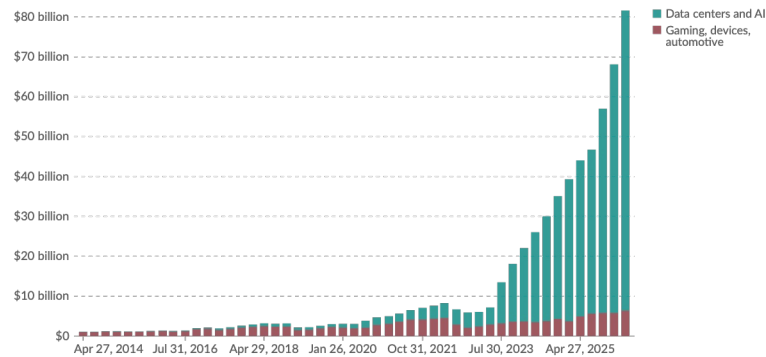
<https://ourworldindata.org/ai-timelines>

Artificial Intelligence and Society

Economic growth

NVIDIA's quarterly revenue by market segment

Quarterly revenue of NVIDIA Corporation across its main market segments, reported in US dollars. NVIDIA designs graphics processing units (GPUs), originally built for gaming and now also widely used to train and run AI models. This data is not adjusted for inflation.



Data source: NVIDIA Corporation (2026)

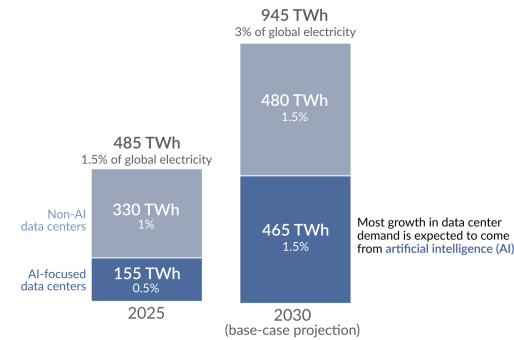
Note: "Data centers and AI" includes GPUs, networking, server CPUs, software, and services sold to data-center customers. "Gaming, devices, automotive" includes revenue from gaming, professional visualization, automotive tech, and chips and technology sold or licensed to other companies.

OurWorldInData.org/artificial-intelligence | CC BY

Energy Consumption

How much of global electricity is used for data centers?

Measured in terawatt-hours (TWh). Estimates for 2025; base-case projections for 2030 from the International Energy Agency (IEA).



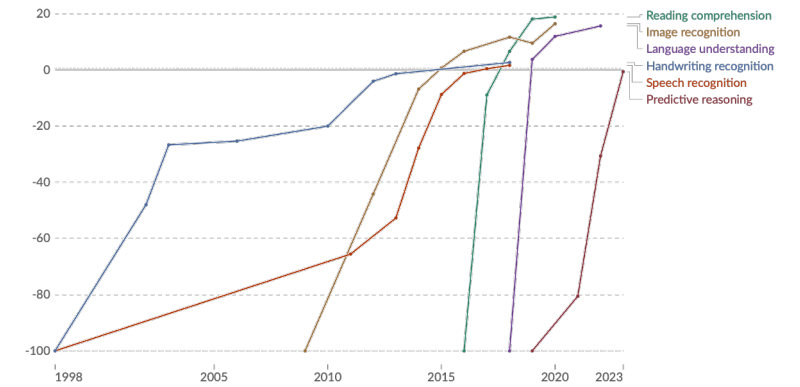
Note: Projections for 2030 are very uncertain, but shown here to reflect expectations that most data center growth will come from AI-focused ones. Data source: IEA (2026). Key Questions on Energy and AI.

CC BY

Benchmark against human performance

Test scores of AI systems on various capabilities relative to human performance

Within each domain, the initial performance of the AI is set to ~100. Human performance is used as a baseline, set to zero. When the AI's performance crosses the zero line, it scored more points than humans.



Data source: Kiela et al. (2023)

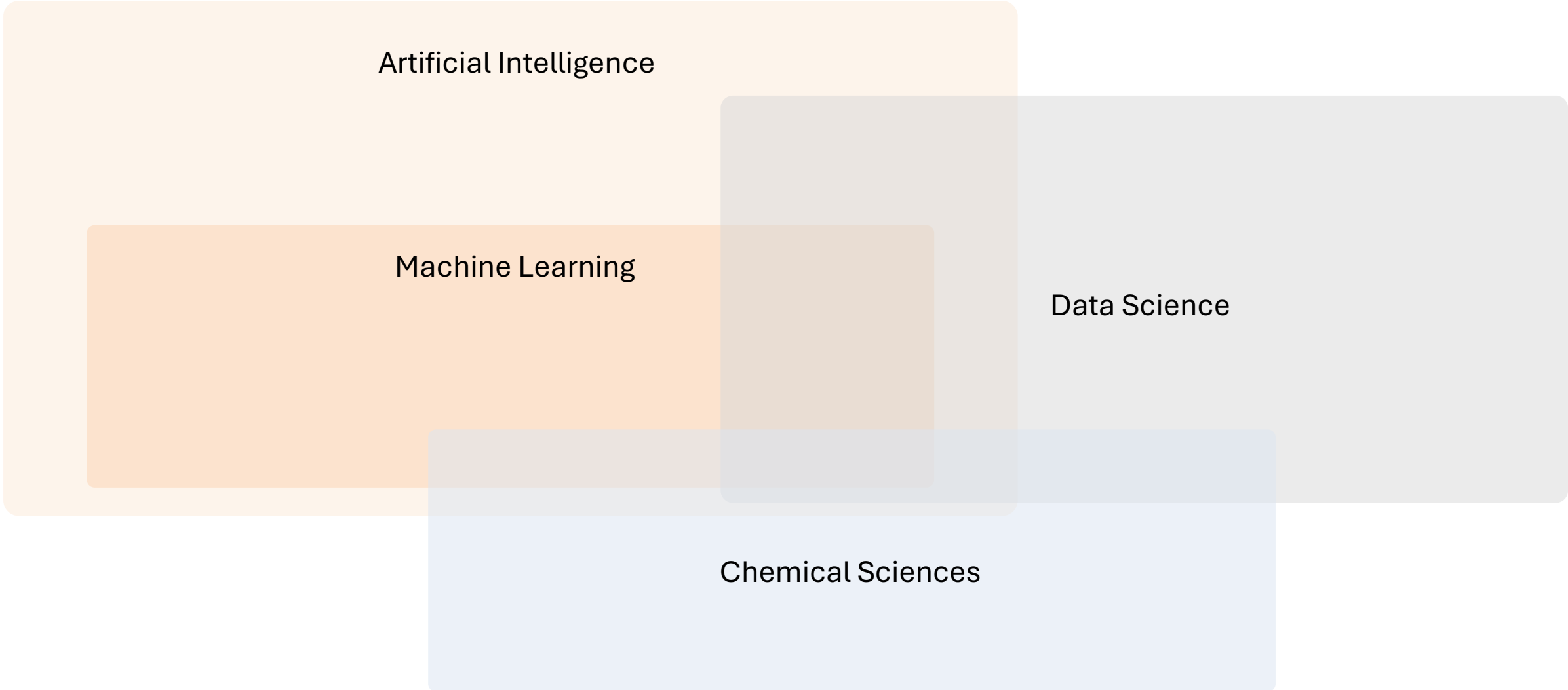
Note: For each capability, the first year always shows a baseline of ~100, even if better performance was recorded later that year.

OurWorldInData.org/artificial-intelligence | CC BY

<https://ourworldindata.org/artificial-intelligence>

<https://ourworldindata.org/how-much-energy-do-data-centers-and-artificial-intelligence-use>

Artificial Intelligence & Machine learning



A Venn diagram illustrating the relationship between four fields: Artificial Intelligence, Machine Learning, Data Science, and Chemical Sciences. The diagram consists of four overlapping rounded rectangles. 'Artificial Intelligence' is a large light orange rectangle on the left. 'Machine Learning' is a smaller, darker orange rectangle nested within 'Artificial Intelligence'. 'Data Science' is a large light gray rectangle on the right, overlapping with both 'Artificial Intelligence' and 'Machine Learning'. 'Chemical Sciences' is a light blue rectangle at the bottom, overlapping with 'Machine Learning' and 'Data Science'. The intersections of these fields are represented by darker shades of their respective colors.

Artificial Intelligence

Machine Learning

Data Science

Chemical Sciences

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Bootcamp Team

J. SCHLEINITZ

Bootcamp lead & organizer



S. WAHLER

Supervised Learning · LLM
Assistants



M. MURPHY

MLIPs & Graph Neural
Networks



C. LIU

Diffusion & Generative
Models



M. YUSOV

Model Interpretability



A. SCHMIDT

Design of Experiments ·
Data Stewardship



B. REINER

Agentic AI



A. GURAJAPU

Hackathon Facilitation,
MLIPs



H. JUNG

Tutorials Preparation



R. FOX

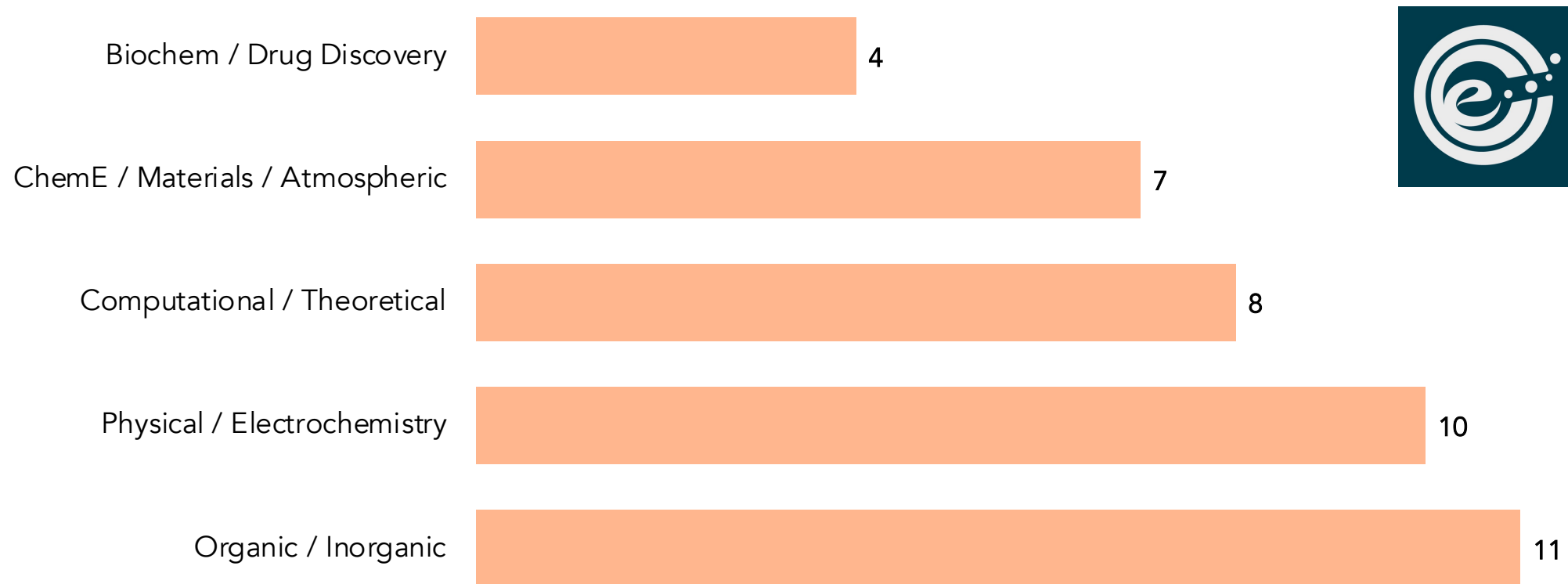
Logistics



T. KANO

Logistics

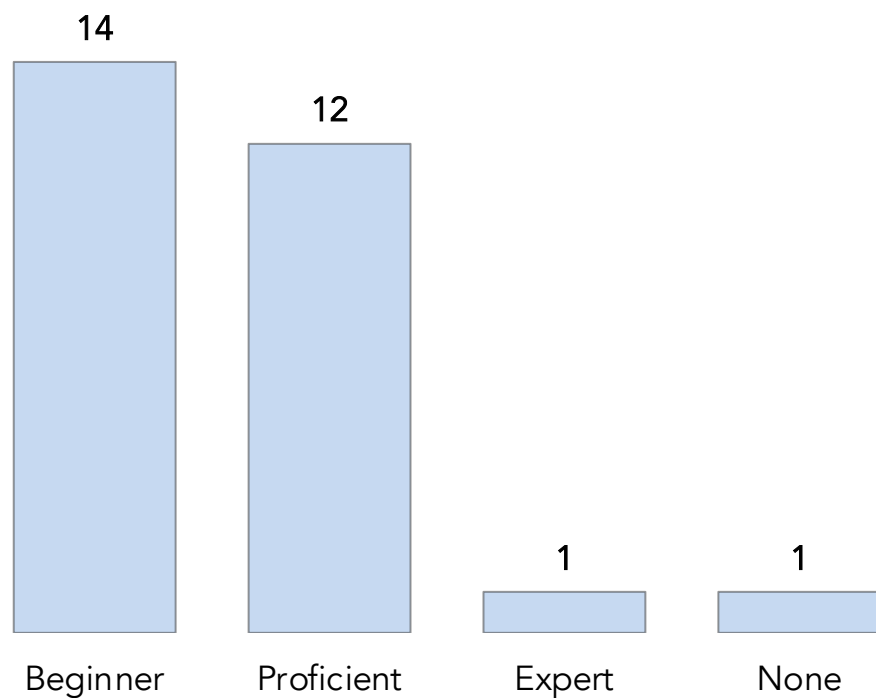
Attendance to the bootcamp



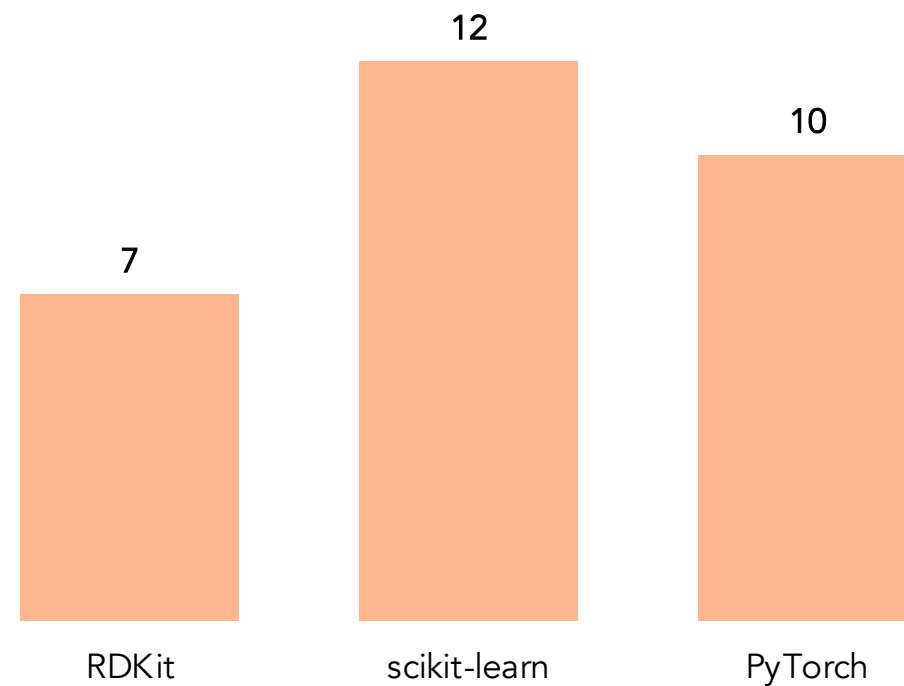
Classification done by Claude based on your responses to the Questionnaire

Coding Experience

Python experience

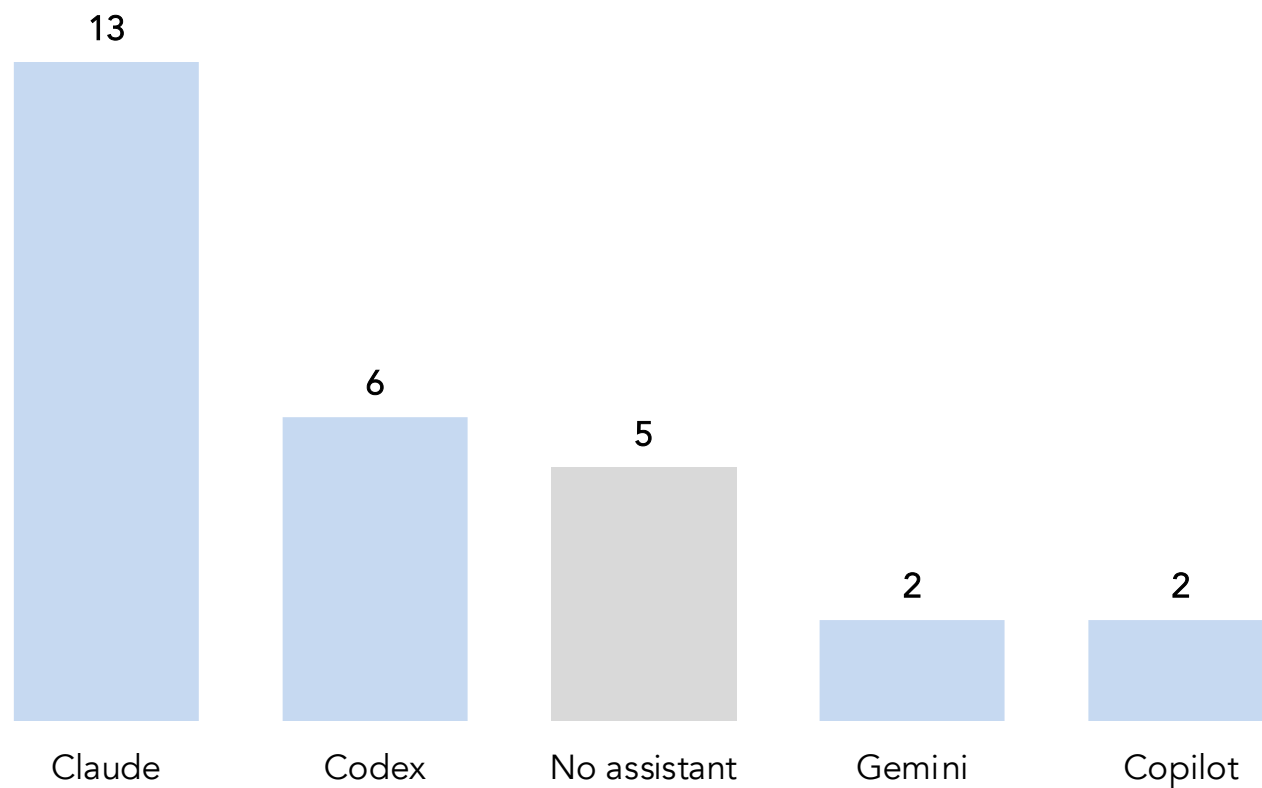


Familiarity with chemistry/ML tools



Based on Bootcamp Questionnaire

LLM assistant usage



~80%

already use an AI coding assistant day to day.

We will discuss how to use them efficiently

Based on responses to the Questionnaire

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Your expectations

"Automate the literature review & summarization"

"Machine-learned interatomic potentials (MLIPs) for MD"

"Ligand design & reaction-yield optimization"

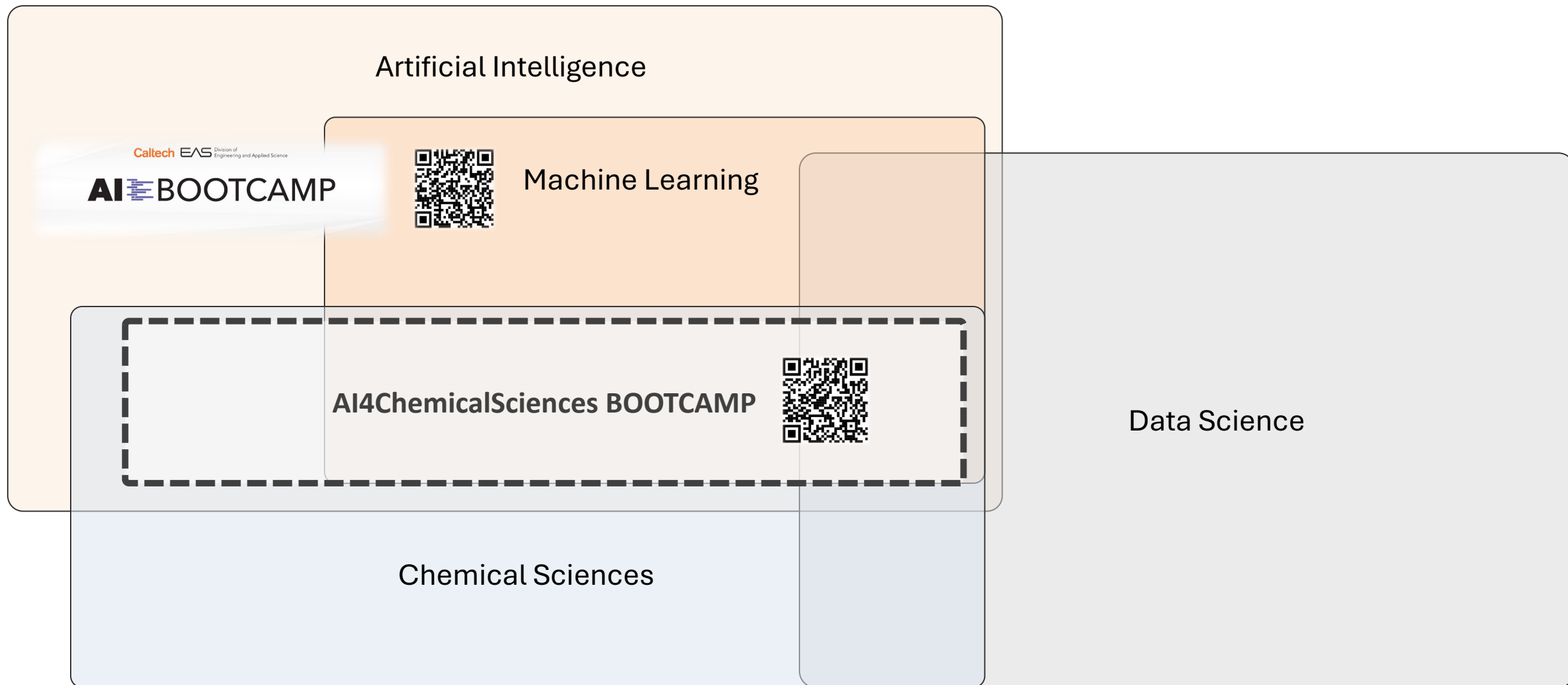
"Understand the "back-end" — which algorithm, why, how it's validated"

"Docking, virtual screening & binding-pose prediction"

"Multi-agent workflows / model selection guidance"

Based on Bootcamp Questionnaire

AI – ML – Data Science and Chemical Sciences



Building Machine Learning Models – What's needed?

DATA

The measurements and observations that ground everything a model learns.



REPRESENTATIONS

How a molecule or reaction becomes numbers a model can actually use.



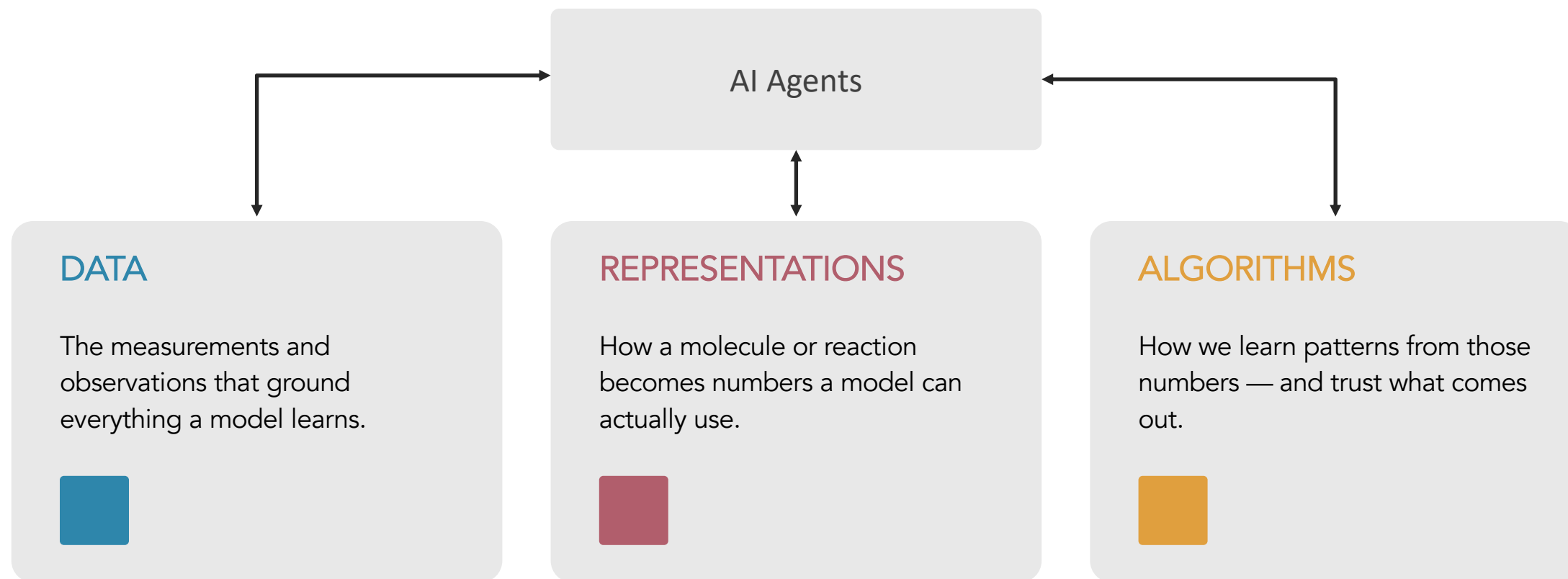
ALGORITHMS

How we learn patterns from those numbers — and trust what comes out.



Models = understanding or black box for optimization

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Agents can help accelerate the integration, they still need guidance

Scientific Data: Acquisition Bottleneck



- Chemical data is expensive and slow to generate especially for wet lab experiments
- Datasets are small compared to vision or language: hundreds to thousands of points, not billions.
- Negative results and failed reactions are rarely recorded — a strong, systematic reporting bias.
- Noise from instruments, batch effects, and inconsistent protocols across labs and instruments.
- Data is scattered across papers, ELNs, and proprietary silos — rarely FAIR or ML-ready.



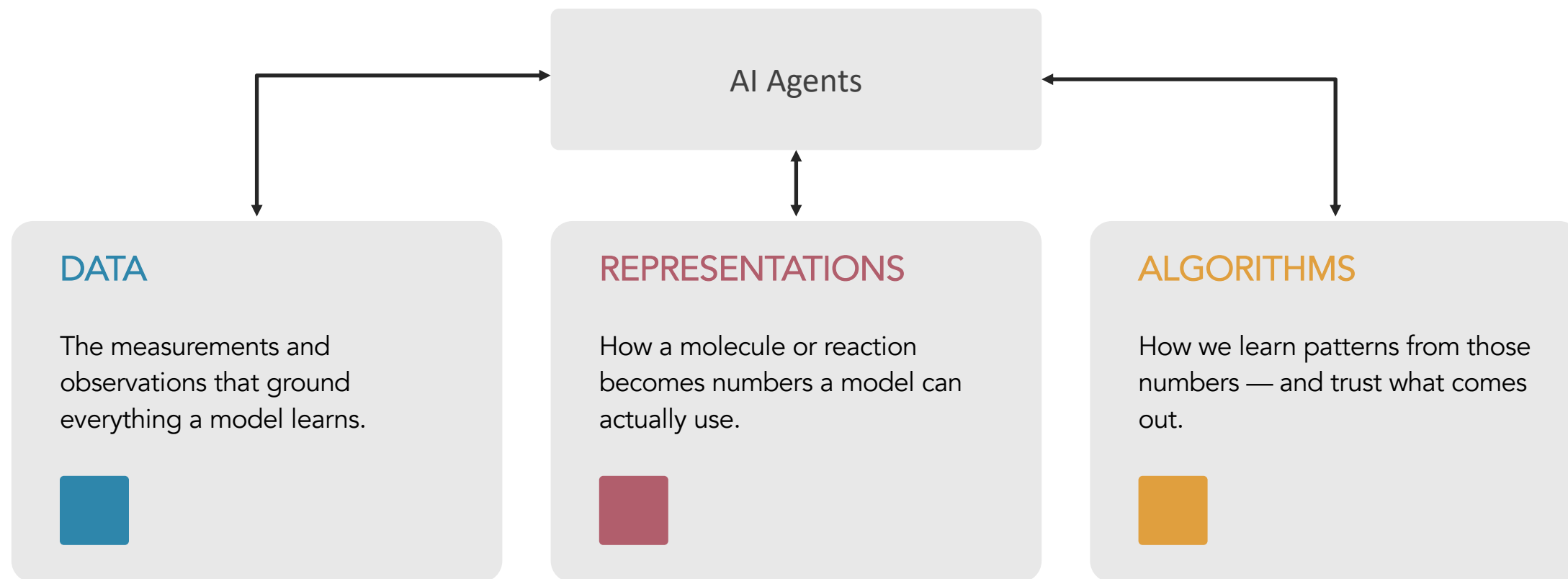
- **Molecules aren't naturally vectors** — there is no single canonical coordinate system for them.
- One molecule, many encodings: SMILES, InChI, SELFIES, fingerprints, 3D graphs — each keeps different information and hides different blind spots.
- Symmetries matter: permutation of atoms, rotation/translation in 3D, stereochemistry, tautomers.
- A representation that throws away the wrong information caps your model before training even starts.

Algorithms: choosing the model

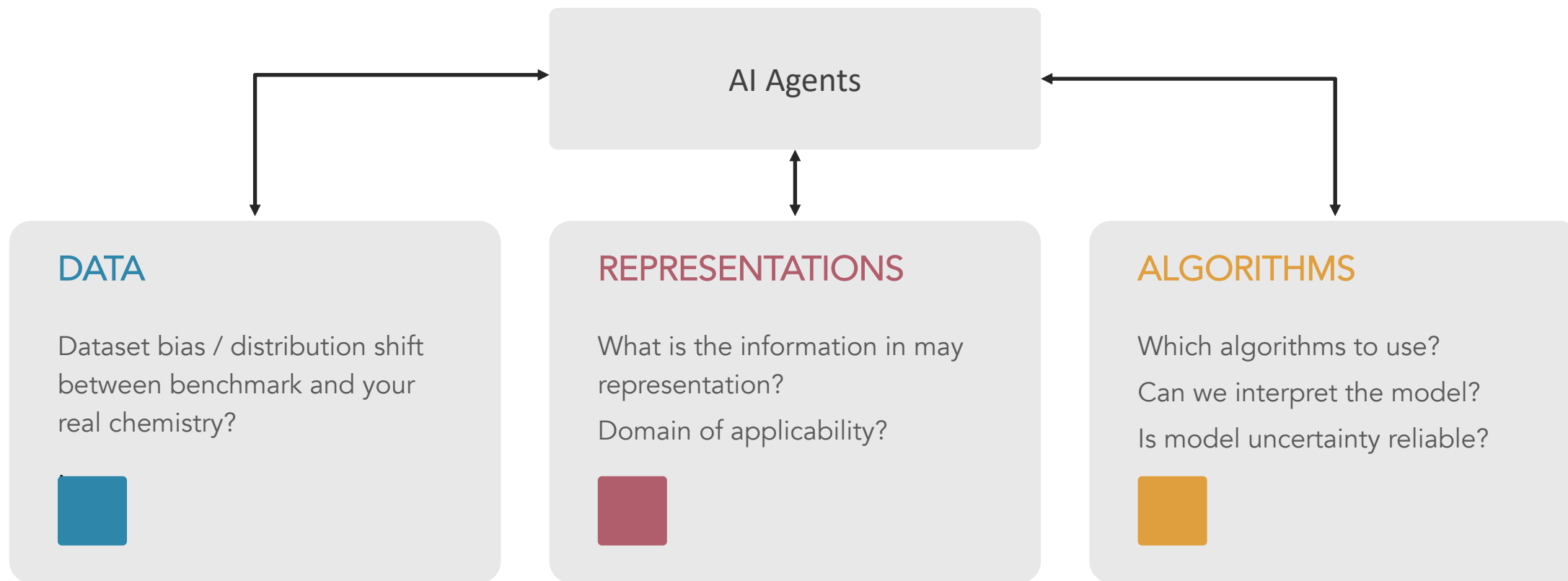


- Which model fits your data: a linear model, a random forest, a graph neural net, a transformer?
- Which model fits your task: extrapolation, interpretability...?
- What model to use to rely on predicted uncertainty?

AI4Chemical Sciences



AI4Chemical Sciences: Learning Objectives



Bootcamp Roadmap

WEEK 1

Aug 10 – 14



Foundations of ML & Representations

Supervised learning, neural networks, molecular representations, GNNs / MLIPs, LLM assistants, generative models.

WEEK 2

Aug 17 – 21



Dataset Design & LLMs

Interpretability, design of experiments, Bayesian optimization, active learning, data stewardship, transformers, agentic AI.

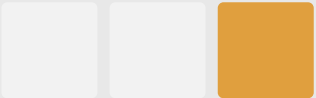
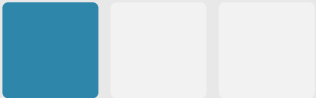
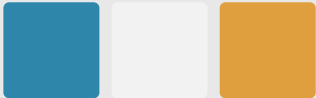
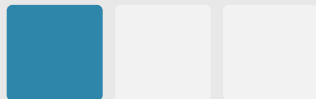
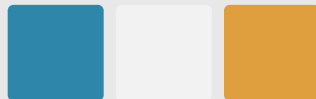
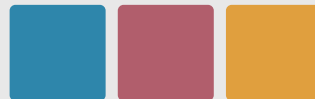
Week 1 — Foundations of ML & Representations



MON	TUE	WED	THU	FRI
9:00 Intro & Logistics   	9:00 Supervised Learning   	9:00 MLIPs — GNNs   	9:00 LLMs as Chemistry Assistants — RAG   	9:00 Hackathon   
13:30 Evolution of Molecular Representations   	13:30 Neural Networks & Deep Learning   	13:30 Learned Representations   	13:30 Generative Models   	  

Week 2 — Dataset Design & LLMs



MON	TUE	WED	THU	FRI
9:00 Model Interpretability 	9:00 Design of Experiments 	9:00 Active Learning 	9:00 Agentic AI for Chemistry 	9:00 Hackathon 
13:30 Data Stewardship 	13:30 Bayesian Optimization 	13:30 Transformers — text-based representations, tokenization 	13:30 Hackathon 	13:30 AI Readiness Wrap-up & Perspectives 

Bootcamp Objectives

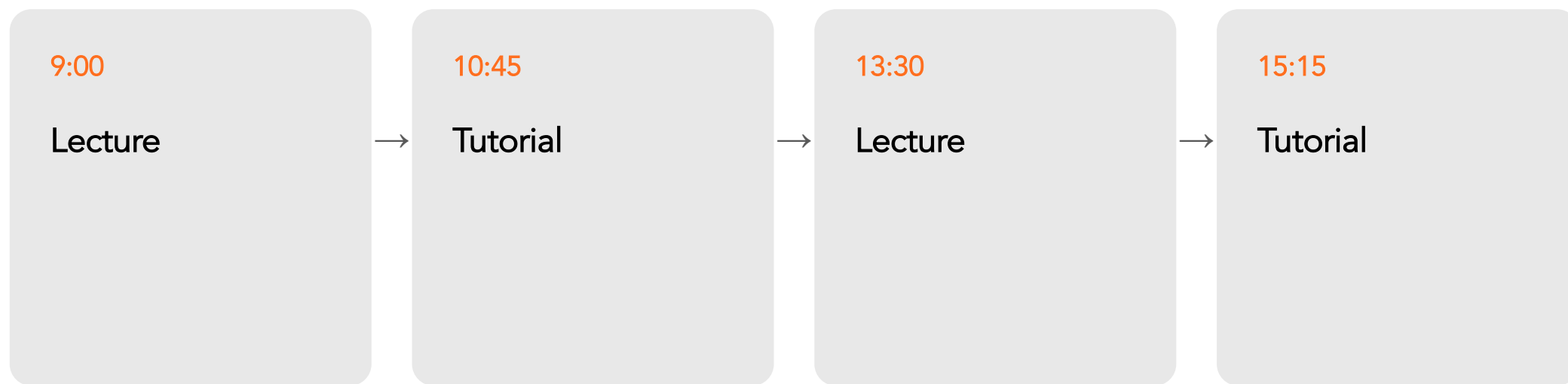
Goal for the two weeks:

- Basic literacy across data, representations, and algorithms
- Basics of when to start/not start an AI/ML project.
- Basics of how to get started coding wise.
- Build a network of chemists interested in AI4Science.

Tutorials objectives

- **Reproducible pipelines:** fixed seeds, versioned data splits, logged hyperparameters.
- Define **baselines**.
- **Code architecture that separates data, features, model, and evaluation:** test components independently.
- Readable code today is debuggable code tomorrow — including code an agent wrote for you.

Lecture are paired with Tutorial



Coffee 10:30–10:45 · Lunch 12:15–13:30 · Coffee 15:00–15:15

Fridays are full Hackathon days — apply the week's concepts to a real chemistry challenge with a presentation at the end.

Bootcamp Objectives

Goal for the two weeks:

- Basic literacy across data, representations, and algorithms
- Basics of when to start/not start an AI/ML project.
- Basics of how to get started coding wise.
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Coding agents are getting stronger, fast

↑ Capability

Agents now scaffold pipelines, debug, and write whole tutorials' worth of code in minutes.



↑ Cost

That capability is not free — token and compute cost scale with how much you let an agent run unsupervised.

↑ Need for direction

More power means you need to specify intent precisely and monitor what actually got done.

~80%

of you already reach for Claude, Codex, Gemini or Copilot.

How to direct them well? How to keep track of what they produce?

Going faster without losing control

- Thursday, Week 2 — Agentic AI for Chemistry: ReAct, Coscientist, multi-agent workflows, self-driving labs.
- Build a Chemistry ReAct agent with PubChem lookup, RDKit tools, and a BO surrogate, with guardrails.
- Every hackathon: use agents deliberately — know what “good” output looks like before you delegate to one.
- Ground rule for the two weeks: review the diff, don't just trust the green checkmark.

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Where to find everything

Schedule



The live day-by-day plan — this page. Subject to small changes as the bootcamp goes.

Lectures

Slides & notes for every lecture, linked directly from the schedule.

Tutorials

Every hands-on notebook — the same ones you'll open each session.

Resources

Curated courses, papers & tools by topic — from ML fundamentals to self-driving labs.

Claude and Colab Licenses

Claude Pro

If you don't have access, send me an email at jul@caltech.edu now

Colab Pro

If you don't have access go to this web page using the QR code, select 100 compute units and raise your hand, Tomomi will help for the payment



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Compute units expire after 90 days.
Purchase more as you need them.

Checking Colab Interface

← → ↻ julschleinitz.github.io/ai4chemistry-bootcamp/tutorials/molecular-representations.html ☆ 📄 🗂️ ⬇️ J ⋮

AI 4 Chemical Sciences Schedule Lectures **Tutorials** Resources

TUTORIAL 1

Molecular Representations

Build, visualize, and compare five molecule-to-vector representations — physicochemical descriptors, Morgan fingerprints, an xtb solvation-energy descriptor, a ChemBERTa foundation-model embedding, and molecular graphs — using the ESOL solubility dataset.

📅 August 10, 2026 · 15:15 – 16:45 ⌚ 90 min <> Python · Google Colab



< Back to schedule

Open in Google Colab
The notebook has most of the code pre-filled. Complete the exercises marked **Exercise** or your code here as you go.

🔗 **Open Notebook**

Let's meet each other

Next up, 10:45 — round-table introductions: your research background and what you're hoping to get out of these two weeks.

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